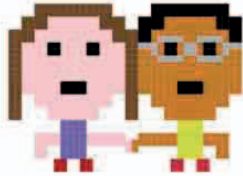


## Loop variables

A loop variable counts the number of times a loop has repeated itself. It starts at the first value in the range (0) and stops one before the last value.



The loop variable

The loop repeats ten times

```
for i in range(10):
    print(i, end=' ')
```

Python stops counting one before the last value

```
>>> 0 1 2 3 4 5 6 7 8 9
```

### △ Simple loop variable

Here, the loop's range doesn't state what the starting value should be. So Python starts counting from 0, the same way as it does with strings.

This tells the program to count in twos

```
for i in range(2, 11, 2):
    print(i, end=' ')
```

```
>>> 2 4 6 8 10
```

The output appears in twos

### △ Counting in twos

This loop has a third value in its range, which tells the loop to count in twos. It stops at 10, which is one loop before the loop variable gets to 11.

This tells the program to count backward

```
for i in range(10, 0, -1):
    print(i, end=' ')
```

```
>>> 10 9 8 7 6 5 4 3 2 1
```

### △ Counting backward

This time the program counts backward from 10, like in a rocket launch. The loop variable starts at 10 and takes steps of -1 until it reaches 1.

## Nested Loops

Loops inside a loop are called "nested loops". In nested loops, the outer loop only repeats after the inner loop has gone round its required number of times.

To make the loops repeat "n" number of times, the last number in the range must be "n + 1"

```
n = 3
for a in range(1, n + 1):
    for b in range(1, n + 1):
        print(b, 'x', a, '=', b * a)
```

Outer loop

Inner loop

### △ Loops inside a loop

In this example, each time the outer loop goes around once, the inner loop goes around three times. So in total, the outer loop is executed three times and the inner loop is executed nine times.

This sum will be printed nine times

The value of "b"

The value of "a"

```
>>>
1 x 1 = 1
2 x 1 = 2
3 x 1 = 3
1 x 2 = 2
2 x 2 = 4
3 x 2 = 6
1 x 3 = 3
2 x 3 = 6
3 x 3 = 9
```

First time around the outer loop (the inner loop repeats three times)

Second time around the outer loop

Third time around the outer loop

### △ What happens

The nested loops print the first three lines of the 1, 2, and 3 times tables. The value of "a" only changes when the outer loop repeats. The value of "b" counts from 1 to 3 for each value of "a".